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A low-complexity window-based learning framework for adaptive 6G beam selection using WiSARD

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A R T I C L E I N F O

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A B S T R A C T

Millimeter-wave (mmWave) communication is cornerstone for 6G, but efficient beam selection in dynamic environments makes exhaustive search impractical. Conventional Deep Neural Networks (DNNs) can leverage rich contextual data (e.g., LiDAR, GPS) for high accuracy but are computationally expensive and operate offline, preventing real-time adaptation. Conversely, existing low-latency online algorithms cannot integrate high-dimensional contextual data, creating a critical trade-off between contextual richness and adaptive capability. This paper proposes a low-complexity framework to bridge this gap, aiming to deliver both high contextual accuracy and rapid adaptation on resource-constrained edge devices. We propose a solution based on a Weightless Neural Network (WNN), the RAM-based WiSARD model, combined with a novel window-based learning strategy that enables adaptation by forgetting obsolete statistics. The framework was evaluated using multimodal (LiDAR and GPS) contextual data against state-of-the-art DNN baselines. In the adaptive window-based evaluation, the WiSARD model achieved a top-1 accuracy of 61.0%, outperforming the baselines (56.0% and 49.0%) while requiring 47% less training data. Critically, the WiSARD model's training time was 99.78% faster than baseline models. The findings establish the window-based WiSARD framework as an effective and practical solution for adaptive 6G beam management.

1. Introduction

Millimeter-wave (mmWave) communication, operating at frequencies between 30–300 GHz, provides massive bandwidth and high data rates, positioning it as a cornerstone of 5G and an enabler for future 6G networks [1]. However, mmWave propagation is highly susceptible to environmental dynamics, suffering from severe path loss, angular dispersion, delay spread, and frequent blockages [2]. This sensitivity to rapid environmental changes constitutes one of the most critical challenges in mmWave systems.

At these high frequencies, the short wavelengths enable the strongly directional transmission required to overcome severe propagation loss. Consequently, minor movements by user (e.g., vehicles, handheld devices), dynamic obstacles (e.g., other vehicles or pedestrians), or environmental variations (e.g., rain, foliage) can significantly impact channel usage within milliseconds. To compensate for high path loss and maintain link reliability, transmitters and receivers employ large antenna arrays to perform beamforming. This technique generates and directs narrow, high-gain beams to establish an aligned, high-quality link between the communicating devices. The task of identifying the optimal transmit-receive beam pair that maximizes the signal-to-noise ratio (SNR) or throughput defines the beam selection problem [3]. In this work, we specifically address the beam selection challenge during the Initial Access (IA) phase. Unlike beam tracking approaches that rely on active link history, our method utilizes environmental context to accelerate the link establishment for users with no prior channel state information.

Traditional beam search strategies (e.g., exhaustive or hierarchical) are robust but computationally expensive and unsuitable for fast-varying mobile environments [2,4]. The short channel coherence time and abrupt blockages require near-instantaneous beam re-selection.

Data-driven methods improve efficiency. Compressed sensing exploits channel sparsity, while supervised and reinforcement learning predict beam indices using contextual data [5–7]. Context-aware multimodal learning further reduces overhead by incorporating auxiliary data like LiDAR, GPS, and camera feeds [8,9].

Despite these advances, state-of-the-art models (DNNs, CNNs) typically operate in an offline batch mode, requiring large static datasets and retraining, which prevents real-time edge operation. While online learning paradigms [10] adapt incrementally, they often use

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Table 1
Performance comparison of ML techniques from the literature.

Ref	Input	ML model	Antenna config.	Dataset	Accuracy Top-1	Top-10
[5]	GPS	Linear SVM	UPA 4X4	s000	33.2%	–
		AdaBoost			55.0%	–
		Decision tree			55.5%	-
		Random Forest			63.2%	–
		DNN			63.8%	–
		DQN			67.5%	–
[11]	LiDAR	3D CNN	-	s000	63.3%	95.0%
[23]	GPS, LiDAR	ViT model	-	s008-s009	59.8%	92.3%
[22]	GPS, LiDAR	MLP + CNN	ULA 32X8	s008-s009	58.0%	92.0%
[19]	GPS, LiDAR	Federated	-	s008-s009	50.0%	91.2%
[13]	GPS, LiDAR	NN		s008-s009	59.5%	92.2%
[8]	GPS, LiDAR, img	DNN	ULA 32X8	s008-s009	56.2%	91.1%
[9]	GPS, LiDAR, img	CNN	ULA 64X8	s008	22.0%	65.0%

gradient-based optimization or deep structures, introducing latency unsuitable for edge constraints.

Furthermore, this creates a trade-off: online approaches lack the ability to process rich, high-dimensional context (e.g., LiDAR), while batch-learning algorithms that can process this kind of data lack real-time adaptability.

Our proposed window-based learning approach bridges this gap by combining a context-aware strategy to capture channel dynamics with the Weightless Neural Network (WiSARD) to efficiently integrate high-dimensional context. We used the ray-tracing technique to evaluate the performance model. The Raymobtime dataset collects realistic datasets of wireless communications, and unimodal (coordinates or LiDAR data) and multimodal (coordinates + LiDAR) information was used for the beam selection problem. We evaluated 3 types of windows to evaluate the performance model and compare the WiSARD results against baseline models.

The proposed framework demonstrates up to a 4.6% improvement in top-1 beam prediction accuracy compared to the best baseline machine learning methods, while simultaneously achieving a 99.98% reduction in training time. These results highlight the potential of the WiSARD-based approach for delivering effective, adaptive, and low-latency beam management in 6G systems.

The main contributions of this paper are:

- Algorithmic contribution: A WiSARD-based algorithm for beam selection offering competitive accuracy with significantly reduced computational complexity.
- Methodological contribution: A sliding window, context-aware approach to capture channel dynamics while integrating high-dimensional context.
- Empirical contribution: Simulation experiments using fixed and incremental window configurations to assess accuracy and efficiency against batch-learning baselines.

The remainder of this paper is organized as follows. Section 2 reviews existing batch and online beam selection approaches. Section 3 presents the system model. Section 4 describes the proposed WiSARD-based learning framework. Section 6 details the proposed methodology for window-based adaptation. Section 7 presents the experimental setup, and discusses the corresponding results. Finally, Section 8 concludes the paper.

2. Review of beam selection

Millimeter-wave (mmWave) systems rely on high-frequency bands (30–300 GHz) to deliver the capacity and low latency required by 5G and 6G networks. While advanced beamforming is essential to overcome inherent path loss and poor penetration, the primary challenge is quickly selecting the optimal beam pair in highly dynamic and non-stationary scenarios.

Modern beam selection research has evolved beyond exhaustive search, yielding approaches that reduce training overhead while maintaining high accuracy. Hierarchical methods, such as those in IEEE 802.11ad/ay, refine beam direction to minimize search complexity, while compressed sensing exploits mmWave channel sparsity to estimate dominant paths. Learning-based approaches predict optimal beams using side information or past observations; these range from supervised and reinforcement learning to lightweight online models for real-time environments. Context-aware strategies further reduce signaling overhead by incorporating non-RF data, such as LiDAR [11–14], GPS [5,7,15–18], and multi-modal combinations [8,9,19–25]. Reported high accuracy often relies on complex models, like Deep Neural Networks (DNNs) and Convolutional Neural Networks (CNNs), operating in batch learning mode.

To provide a quantitative basis for comparison, Table 1 compiles the Top-1 and Top-10 accuracy results from representative batch approaches, highlighting the performance achievable with rich contextual data from the Raymobtime dataset.

However, a critical bottleneck arises: while DNNs demonstrate high accuracy, these models incur high computational costs and demand large datasets. If we consider the highly dynamic nature of the mmWave environment (e.g., mobile users, fast blockages), relying on resource-intensive batch learning (requiring periodic retraining and data collection) is an unsuitable strategy for real-time beam adaptation and low-latency services.

This high computational demand and the inability to adapt immediately to non-stationary channel conditions motivate a fundamental shift toward true online learning strategies for beam selection.

Online learning, which incrementally learns from data streams, is well-suited for dynamic 6G requirements like URLLC (Ultra-reliable low-latency communication). This paradigm, primarily explored via Multi-Armed Bandit (MAB) and Deep Reinforcement Learning (DRL) frameworks, is inherently better suited for continuous adaptation. DRL techniques, for instance, use GRUs to capture temporal dependencies [26], hierarchical DRL for joint beam optimization [27], and simplified DQNs to reduce computational weight [28].

The MAB framework is preferred for extremely low complexity and fast convergence, often achieving near-optimal performance in V2X scenarios [10]. MAB extensions handle THz directionality (C-MAB) [29], channel non-stationarity (Time-Varying MAB) [30], and network-level optimization like JPBS [31] or decentralized multi-agent allocation [32].

While reinforcement learning offers adaptability, other recent frameworks focus on temporal feature extraction. For instance, Liu et al. [33] proposed the SA-TBP framework, which utilizes speed-adaptive temporal sequence modeling to predict future beam states based on uses mobility patterns. In contrast to such sequence-based architectures which require capturing temporal dependencies effectively, our work approaches adaptability through a ‘sliding window’ retraining mechanism. Instead of modeling the time-series explicitly, we update the semantic memory of a Weightless Neural Network (WiSARD) by continuously shifting the training data horizon, providing a computationally efficient alternative for dynamic environments

Parallel to temporal modeling, distributed learning paradigms have been explored to mitigate the overhead of centralized training. Ali and Koucheryavy [34] recently introduced a Clustered Federated Learning approach for adaptive beam tracking in 5G/6G networks, allowing user clusters to collaboratively learn beam patterns. While effective for shared knowledge, federated strategies incur communication overhead for model aggregation. In contrast, our proposed WiSARD framework prioritizes standalone, low-latency operation. By performing lightweight retraining locally on the user equipment (or edge node) without the need for gradient exchange, we achieve rapid adaptation with minimal signaling overhead.

Despite these advancements, a critical research gap persists: existing Online Learning solutions do not effectively integrate rich, high-dimensional contextual information (LiDAR/Images). DRL and MAB proposals rely on low-dimensional context (GPS, DOA, SNR) to guarantee low latency. Hybrid approaches that use DNNs for feature extraction, such as Deep Contextual Bandits (DCB) [35] or Deep Transfer Learning (DTL) [36], introduce computational burdens unsuitable for resource-constrained edge devices.

Indeed, efficiency is a critical constraint across all layers of mmWave system design. Recent studies, such as López-Benítez and Alhulayil [37], have explicitly quantified the computational burden of fading models, demonstrating that complex channel estimations can bottleneck real-time performance. This same computational constraint applies to beam selection: high-complexity models cannot effectively operate within the tight latency bounds of 6G networks.

This leads to a clear trade-off: Batch learning provides Contextual Richness at the expense of adaptability and Low Latency; Online learning provides Adaptability and Low Latency at the expense of Contextual Richness.

Our proposed framework, using the WiSARD model and a novel window-based adaptation strategy, bridges this gap. The RAM-based WiSARD model offers inference latency orders of magnitude lower than DNNs and efficiently processes high-dimensional features (e.g., LiDAR), achieving high contextual accuracy without computational overhead. The window-based approach provides lightweight online adaptation, allowing the model to forget obsolete channel statistics and adapt to the non-stationary 6G environment. This work, therefore, presents an innovative solution delivering high contextual accuracy at the low latency required for 6G URLLC and edge deployment.

2.1. Benchmark models

As this study represents an intermediate step between batch and true online learning, we evaluate how models originally designed for batch learning perform under a window-oriented learning approach. To ensure a fair comparison between machine learning techniques, we selected two studies that used the same dataset, antenna configuration, and contextual information, and that reported strong performance metrics—making them suitable for benchmarking.

Specifically, the models proposed by J. Ruseckas et al. [22] and B. Salehi et al. [8] were adopted as baselines for evaluating our proposed approach. From now on we will call them Batool and Ruseckas models. For the LiDAR input, both authors employed ResNet-inspired architectures. However, their implementations differ in several design aspects, as summarized in Figs. 1 and 2. These figures provide a comparative overview of the models, highlighting architectural components such as the type and number of layers, neurons, and other structural details.

3. System model

This paper aims to investigate an efficient method for selecting the optimal pair of transmit and receive beams that maximize the received signal power in a mmWave MIMO communication system. In this section, we formally present the system model, key assumptions, and the definition of the beam prediction problem.

We consider a mmWave multiple-input multiple-output (MIMO) communication system in which a base station (BS), equipped with N_t antennas arranged in a Uniform Linear Array (ULA), communicates with a user equipment (UE) equipped with N_r elements ULA. The inter-element spacing in both arrays is assumed to be $\lambda/2$, where λ is the wavelength of the carrier frequency. Given that the mmWave propagation environment is typically characterized by a limited number of dominant paths, the channel is assumed to be spatially sparse and is modeled according to the geometric Saleh-Valenzuela model [38] as:

$$\mathbf{H} = \alpha_l \mathbf{a}_R(\theta_l^R)\mathbf{a}_T^H(\theta_l^T), \tag{1}$$

where L is the number of propagation paths, α_l represents the complex gain of the lth path, $\mathbf{a}_T(\theta_l^T)$ and $\mathbf{a}_R(\theta_l^R)$ denote the transmit and receive steering vectors, respectively, associated with the angles of departure (AoD) and arrival (AoA). For a ULA with N elements, the array response vector for an angle θ is expressed as:

$$\mathbf{a}(\theta_k) = \frac{1}{\sqrt{N}}\left[1, e^{j\pi\sin(\theta)}, \dots, e^{j\pi(N-1)sin(\theta)}\right]^T \in \mathbb{C}^{\mathbb{N}} \tag{2}$$

The received signal after transmit and receive beamforming can be described as:

$$y_{(t,r)} = \boldsymbol{w}_r^H \boldsymbol{H} \boldsymbol{w}_t s + \boldsymbol{w}_r^H n \qquad , \tag{3}$$

where $\boldsymbol{w}_t \in \mathbb{C}_{\mathbb{t}}^{\mathbb{N}}$ and $\boldsymbol{w}_r \in \mathbb{C}_r^{\mathbb{N}}$ are the transmitter and receiver beamforming vectors, respectively, S is the transmitted symbol, and n is the additive noise vector.

This model highlights that the received signal $y(t,r)$ depends jointly on the channel characteristics and the choice of the beamforming vectors, which are determined by the precoder and combiner design.

In practical mmWave systems, the set of possible beamforming vectors is limited to predefined codebooks at the transmitter and receiver, denoted as:

$$\boldsymbol{C}_t = \left\{\boldsymbol{w}_{t_1}, \dots, \boldsymbol{w}_{t|C_t|}\right\}, \qquad \boldsymbol{C}_r = \left\{\boldsymbol{w}_{r_1}, \dots, \boldsymbol{w}_{r|C_r|}\right\} \tag{4}$$

where codewords are commonly constructed from Discrete Fourier Transform (DFT) vectors or other steering dictionaries that discretize the angular domain. The beam selection problem can thus be formulated as finding the optimal beam pair index i^* corresponding to some (t*,r*) that maximizes the received signal power $p(y_i)$:

$$i^* = \underset{i\in\{1,2,\dots,m\}}{\text{argmax}} \left(P(y_i)\right), \tag{5}$$

where $m \leq |C_t||C_r|$ represents the total number of possible beam pairs.

We considered a transmitter (Tx) with 32 elements of antenna and receivers (Rx) with 8-elements antennas array. This represents 256 possible beam pairs to select the best. Exhaustive evaluation of all $m = 256$ pairs is typically infeasible. It is computationally expensive and unsuitable for highly dynamic environments, such as vehicular mmWave communication, where rapid environmental changes continuously alter the optimal beam directions.

To mitigate this issue, environment-aware beam prediction can significantly reduce the search space. A smaller subset $\boldsymbol{k} \subseteq C_t \times C_r$ of candidate beams is identified with a high probability of containing the optimal pair; subsequently partial beam training is then performed only over $\boldsymbol{k}$. In environments with rapid geometry changes (vehicles, pedestrians, transient blockages), contextual sensors such as LiDAR and positioning (GPS) provide rich spatial information that correlates with dominant propagation directions and potential blockages.

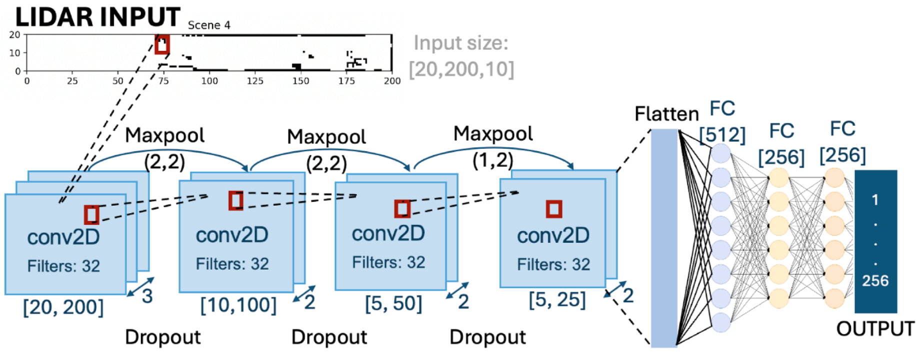

Fig. 1. Neural Network model used in [8] for LiDAR input.

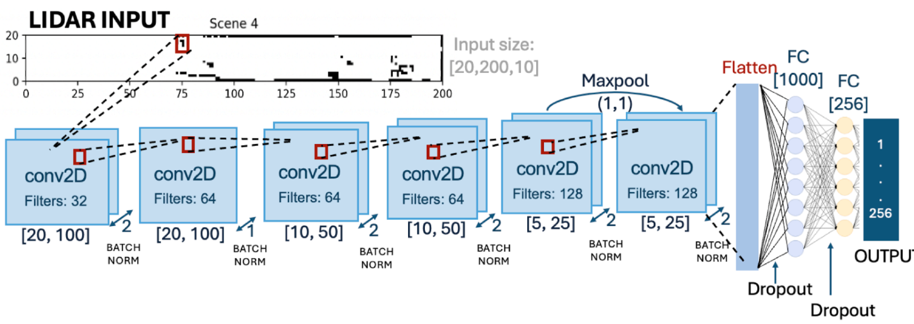

Fig. 2. Neural Network model used in [22] for LiDAR input.

To make operational this context-aided selection, we assume a system architecture supported by a reliable control plane (e.g., Non-Standalone 5G or sub-6 GHz link). The data exchange is defined as follows:

- Signaling: The UE transmits its GPS coordinates to the BS via the control link during the initial connection request. Conversely, LiDAR data is acquired via infrastructure-based sensing (BS-mounted), eliminating the overhead of uplink point-cloud transmission.
- Frequency: This context acquisition occurs once per link establishment attempt (Initial Access), providing the input snapshot required for the WNN to generate the candidate subset $\boldsymbol{k}$.
- Role of Context: The GPS data correlates with the dominant Line-of-Sight (LoS) direction, while the LiDAR point cloud captures the geometry of obstacles and reflectors, aiding in the identification of viable Non-Line-of-Sight (NLoS) paths when LoS is blocked.

This setup ensures that overhead is limited to a single coordinate exchange and local sensor reading per connection attempt. These modalities, therefore, enable informed pruning of the beam search space when fused appropriately with radio measurements.

Motivated by this observation, we propose a machine learning-based approach using a Weightless Neural Network (WNN), specifically the WiSARD model, to predict the most likely subset $\boldsymbol{k}$ of optimal beams based on contextual information, such as spatial or environmental features. Unlike online methods that operate on limited and low-dimensional input data, our proposed approach is designed to handle rich environmental information, such as LiDAR-based features, which provide a detailed spatial description of the surroundings. This allows the system to adapt to dynamic environments and achieve high accuracy in beam prediction while significantly reducing the number of beam measurements required during initial access.

Fig. 3 illustrates the system concept: the BS and UE with their ULAs, the set of candidate beams, the sensed environment (buildings, vehicles, trees), and the context-aware WiSARD predictor that recommends a reduced beam subset for partial training.

4. WNN-based learning method

Weightless Neural Networks (WNNs) represent a class of machine learning models that utilize a memory-based approach for pattern recognition, diverging significantly from conventional weighted networks. Instead of optimizing synaptic weights, WNNs operate by storing and retrieving binary patterns directly within Random Access Memory (RAM) structures [39].

This RAM-based architecture inherently provides high processing speed, low computational complexity, and deterministic inference, making it suitable for real-time applications.

In this section, we present a high-level overview of the WiSARD network architecture and formulate the beam selection problem using this mechanism.

4.1. Fundamentals of RAM-based learning

The *WiSARD* (Wilkie, Stonham, and Aleksander's Recognition Device) model is a prominent WNN based on Random Access Memory (RAM) neurons [40]. The network architecture is composed of M discriminators, where each discriminator D_c is dedicated to learning a specific class c. Each discriminator consists of a set of N RAM nodes with an address size of n bits.

The learning and inference processes rely on a direct mapping between the input pattern and memory addresses:

- **Mapping (Addressing):** The binary input pattern X is partitioned into N tuples of n bits. Each tuple serves as an address to access a specific memory location in a corresponding RAM node.
- **Training Phase (Write):** To train the network on a sample belonging to class c, the system identifies the memory locations addressed by the input pattern within discriminator D_c and sets their content to logical '1'.
- **Inference Phase (Read):** During classification, the network queries all discriminators with the new input pattern. Each discriminator sums the values found at the addressed memory locations. The

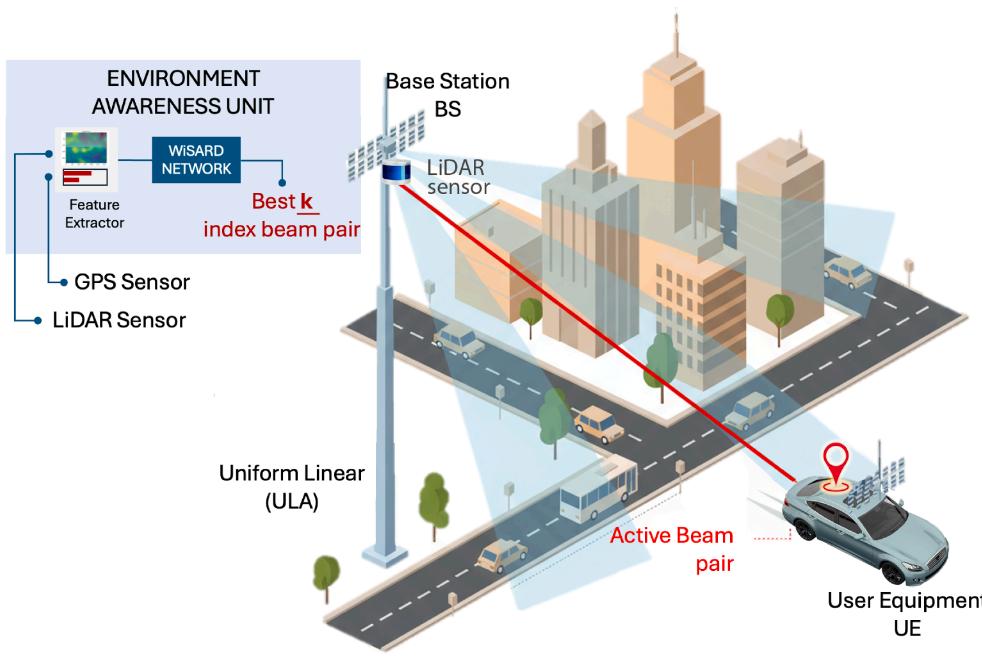

Fig. 3. High level representation of system.

class associated with the discriminator yielding the highest sum (response) is selected as the prediction.

This mechanism allows WiSARD to learn non-linear patterns with $O(1)$ complexity, as training and inference require only memory access operations rather than iterative floating-point multiplications. Furthermore, it eliminates the need for gradient-based updates and is inherently suitable for hardware parallelization.

The configuration of the RAM modules is critical to performance, governed primarily by the *address size* hyperparameter. This parameter dictates the number of input bits mapped to a single RAM, controlling the granularity of learned patterns. A fundamental trade-off exists: smaller address sizes enhance generalization but risk address collisions, whereas larger sizes improve discriminative power at the cost of increased memory consumption. Therefore, this hyperparameter must be carefully tuned to balance memory footprint, precision, and generalization.

The inherent simplicity and low computational cost of the WiSARD model make it highly suitable for resource-constrained applications, such as IoT and embedded systems. In this work, we leverage these properties to evaluate the suitability of WNNs for the beam selection challenge in millimeter-wave MIMO systems. The capability of WiSARD to perform fast, deterministic inference with minimal computational overhead establishes it as a promising candidate for online and real-time decision-making in next-generation cellular networks.

4.2. *WiSARD for beam selection*

We model the beam selection strategy as a multi-class classification problem where the WiSARD network maps environmental context to optimal beam configurations. The specific I/O structure is illustrated in Fig. 3 and defined as follows:

- **Network Input:** The input is a binary vector representing the user's context. The raw coordinates and LiDAR data act as features that undergo a transformation process into intelligible representations suitable for WiSARD models:
  - *GPS Data:* We apply thermometer encoding [41,42] to the normalized coordinates to preserve spatial proximity in the binary domain. This allows the model to handle continuous-valued positional data without losing spatial relationships. A key parameter is the thermometer resolution, which influences the trade-off between input granularity and memory efficiency. A detailed analysis in [43] identified the optimal configuration for this task. Algorithm 1 summarizes the complete encoding procedure.
  - *LiDAR Data:* The point cloud is projected onto a 2D grid, quantized, and flattened into a binary array. These data were obtained from the Raymobtime dataset, where the dimensions are already quantized. In this study, we discard the z-axis to simplify representation and apply a variance thresholding technique [44] to remove features with low variability. This step reduces noise and computational overhead while retaining the most informative features for classification.
- **Network Output:** The network contains M discriminators, where each discriminator corresponds to a unique beam pair index $i \in \{1, \dots, M\}$ from the codebook defined in Eq. (5). The predicted beam is the index of the discriminator with the highest response.

5. Dataset description

The Raymobtime dataset [1] was utilized for experimental simulations in this study. This dataset is widely recognized for its comprehensive multimodal data, a high-fidelity synthetic dataset generated by ray tracing and designed for ML-aided mmWave beam selection. The data are generated at the servers at UFPA and UT Austin hosted by the research groups coordinated by Prof. Aldebaro Klautau, Prof. Robert Heath and Prof. Nuria González-Prelcic.

It has been extensively used in state-of-the-art techniques for beam selection and beam tracking in simulations of wireless communication in vehicular scenarios [5,8,21,22,43,45].

The methodology used to generate the channels is summarized in Fig. 4. The data generation pipeline integrates two distinct simulation environments:

1. Mobility (SUMO): Traffic simulation is managed by SUMO (Simulation of Urban MObility), which generates realistic trajectories for

[1] https://www.lasse.ufpa.br/raymobtime/

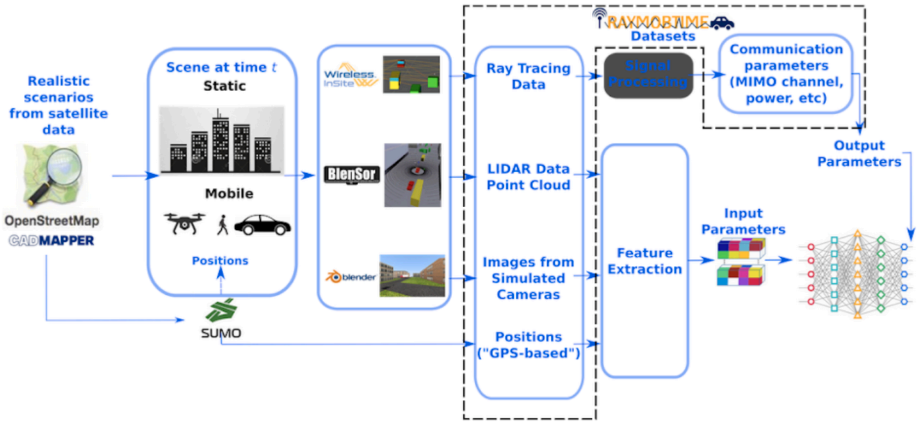

Fig. 4. Raymobtime methodology [46].

Table 2
Raymobtime structure.

Feature Domain	Data Type	Dimension	Description	Archive type
Input (context)	LiDAR	4D structure	[Max number of TX/RX pairs in episode, PCD(X, Y, Z)] PCD(X, Y, Z) - Uniform quantized representation	NPZ
	GPS	9 columns	1. Val-(Channel Valid [V] or Invalid [I]) 2. EpisodeID (Int) 3. SceneID (Int) 4. VehicleArrayID (Number of the Rx in the Vehicle) 5. VehicleName (Example: flow1.2) 6.X (Float - Coordinate) 7.Y (Float - Coordinate) 8.Z (Float - Coordinate) 9.LOS (Line of Sight - True(1), False(0)	CSV
Output (Label)	Beam data	3D structure	[Valids channels, Tx Codebook, Rx Codebook] Describe codebook	NPZ

vehicles and pedestrians, accounting for lane discipline, traffic lights, and inter-vehicle dynamics.

2. Propagation (Wireless InSite): The physical layer is simulated using Remcom Wireless InSite. Ray-tracing is performed at 60 GHz (V-Band) to capture site-specific channel characteristics, including reflection, diffraction, and blockage effects caused by dynamic objects.

We utilize two specific scenarios: s008 and s009 (a canyon-like urban street with moderate traffic). These datasets share similar physical characteristics, with both representing streets in Rosslyn city with 10 mobile receivers, and the frequency used for transmission is 60 GHz.

To visualize the data structure used, the Table 2 presents a Raymobtime data structure. The system inputs include the user's location (GPS) and local geometry (LiDAR point cloud), while the target output is the received power vector across the beam codebook.

In the simulations of Raymobtime, the 'episode' and 'scenes' are concepts used to organize the placement procedure of mobile objects (vehicles, pedestrians, drones, etc.). In the majority of the dataset, an episode is composed of sequential scenes ('snapshot'), but in the s008 and s009 datasets, the episode contains only one scene. It means that the receptors will be the same throughout all simulations of that episode. In a new episode, all scenario changes with no temporal correlation between two consecutive episodes. The gap between two consecutive episodes/scenes is 30 s.

The s008 dataset comprises 2085 episodes, of which 58% of wireless communication is Line-of-sight (LoS) and 42% are Non Line-of-sight (NLOS), while s009 contains 2000 episodes with 15% LoS and 85% NLoS. These numbers highlight the high imbalance of the dataset, reflecting real-world conditions. The average number of receptors per 100 episodes in each dataset is presented in Fig. 5.

6. Window-oriented learning approach

To evaluate the effectiveness of machine learning techniques for beam selection in dynamic mmWave systems, we developed a comprehensive methodology that compares three distinct window-based approaches: Fixed Window, Incremental Window, and Sliding Window. The goal is to identify the most efficient solution for establishing robust links under varying environmental conditions. For this analysis, the window concept is quantified in terms of the number of episodes, where each episode is composed of multiple time-contiguous samples, as shown in Fig. 5.

Fixed Window Approach In this method, a fixed-sized data window W_{fixed} is used to store an initial set of data samples for training. The model is trained once on this fixed dataset and tested with each new event (episode) (Fig. 6). Each episode represents a distinct vehicular connection scenario, characterized by varying environmental and mobility conditions. Accuracy is calculated for each episode, providing insight into the model's generalization capability and performance decay under channel drift, serving as a static baseline.

Incremental Window Approach In this approach, the training dataset grows incrementally over time, accumulating all past events as new data becomes available, W_{fixed}, as represented in Fig. 6.

Unlike the Fixed Window Approach, this method retains the entire history of events, allowing the model to leverage comprehensive knowledge for beam selection. While this approach is impractical in real-world scenarios due to the challenges of maintaining and processing an ever-expanding dataset, it serves as an 'ideal' benchmark in this study. By treating the entire historical dataset as available for training, the Incremental Window Approach provides a theoretical upper bound for performance, offering insights into the maximum improvement achiev-

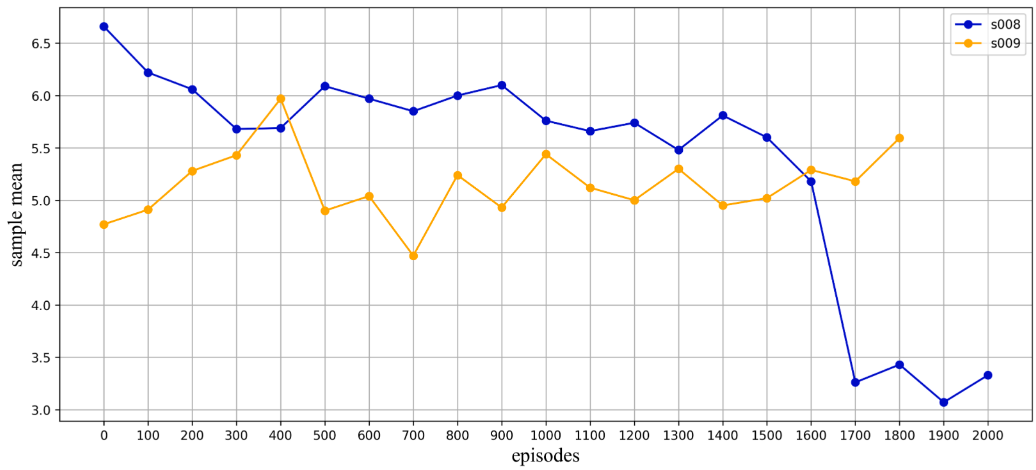

Fig. 5. Samples mean per 100 episode.

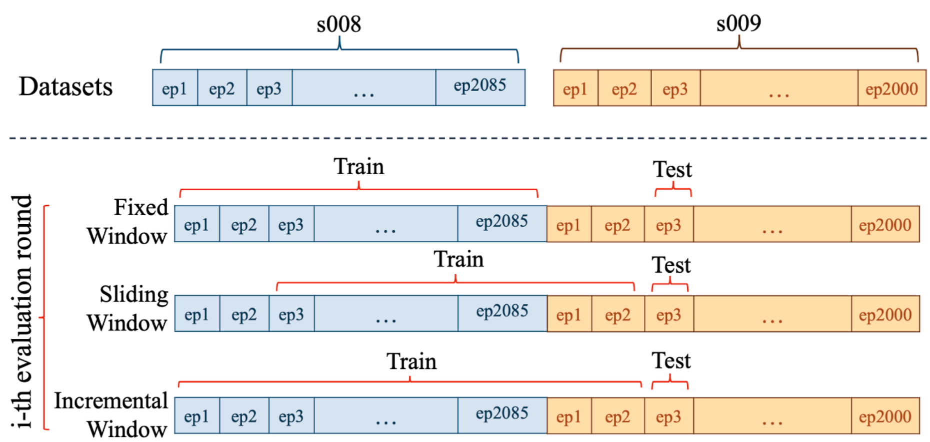

Fig. 6. Diagram of window learning proposal.

able with complete knowledge of past events. The model's accuracy is evaluated after each new episode.

The **Sliding Window Approach** maintains a constant training dataset size. However, the dataset is dynamically updated over time by incorporating the newest episode and simultaneously removing the oldest. This sliding window mechanism allows the model to be trained on the most recent and relevant information, effectively tracking the temporal dynamics and mitigating the effects of concept drift prevalent in the wireless environment. The performance of the machine learning models is evaluated after each new event, being measured continuously. To thoroughly assess the robustness and efficiency of this approach, we varied the size of the sliding window dataset. We selected five different window sizes: 100, 500, 1000, 1500, and 2000 episodes. Fig. 6 illustrates the window mechanism

Temporal Interpretation of Window Sizes To contextualize these window sizes against the environmental dynamics, it is important to note that episodes in the dataset are separated by approximately 30 s. Consequently, a window of $W = 100$ episodes represents a history of ≈ 50 min, capturing short-term variations in traffic density and local blockage patterns.

In contrast, a window of $W = 1000$ episodes covers a duration of ≈ 8.3 h. This larger span allows the model to learn long-term statistical averages of the channel but introduces a higher inertia, potentially slowing down adaptation to significant environmental shifts (e.g., the transition from low-traffic to high-congestion states). This trade-off between stability (long history) and reactivity (short history) is central to the analysis presented in Section 7.3.

To ensure a rigorous evaluation during the transition between the training environment (s008) and the testing environment (s009) shown in Fig. 5, we adopt a prequential protocol where the model is tested on the current episode before it is used for training. This process is formalized in Algorithm 2. This algorithm confirms that performance metrics are calculated solely on unseen data frames, reflecting the system's real-time adaptability.

7. Experimental results

This section provides a comprehensive evaluation of the proposed window-based WiSARD framework. We first detail the data preprocessing techniques employed. We then introduce the baseline machine learning models selected for comparison. Following this, we analyze the performance of the proposed WiSARD model, focusing on its adaptive capabilities. Finally, we present a comparative analysis, benchmarking our approach against the baselines in terms of prediction accuracy and computational requirements.

7.1. Preprocessing data

For the coordinates data, it was necessary to establish two key parameters: The resolution thermometer and the address size memory. In

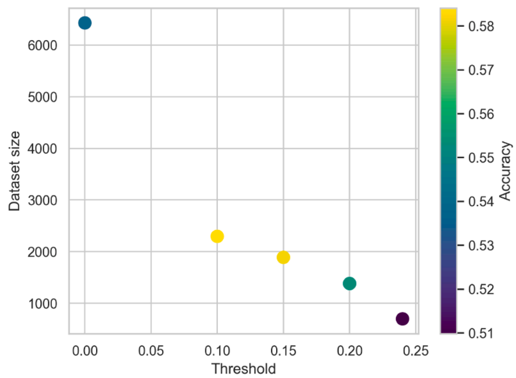

Fig. 7. Performance evaluation of WiSARD model using a variance threshold technique in 2D LiDAR data.

[43], the authors establish a methodology to find these parameters. They evaluated 12 address-size memories, and 10 thermometer resolution values. Accordingly, this work employs a memory address size of 32 bits and a thermometer resolution of 8 bits, which provide a suitable balance between representation precision and computational cost for the WiSARD model.

For LiDAR data, a variance thresholding technique was implemented, and to determine the optimal threshold, five levels were experimentally evaluated based on the WiSARD model's performance. Although the highest classification accuracy (58.4%) was achieved with a threshold of 0.10, a threshold of 0.15 was selected as the final hyperparameter. This choice reduced the dataset size by approximately 17% while resulting in only a 0.39% decrease in accuracy, thus offering a favorable trade-off between performance and data efficiency, as shown in Fig. 7.

7.2. Baseline comparison analysis

In this section, we compare the proposed framework against state-of-the-art baselines using accuracy and training time as performance metrics. The accuracy metric is defined as the proportion of test samples where the predicted beam index i_{pred} matches the ground-truth optimal index $i^* = \text{argmax}_i P(y_i)$.

It is important to note that the Top-1 accuracy metric is strict and does not explicitly account for the angular proximity of the prediction to the target. However, given the spatial structure of the beam codebook, the Top-5 accuracy serves as a metric for geometric "nearness," capturing instances where the model predicts spatially adjacent beams or strong secondary propagation paths.

To ensure a fair evaluation, we implemented the same methodology used in the baseline studies to train and test the models under a batch learning configuration. This approach allowed for consistent measurement and comparison of key performance metrics across all models.

The accuracy metric measures the proportion of correct predictions relative to the total number of predictions. When comparing the baseline models with the proposed WiSARD-based approach, the simulations revealed that, for LiDAR data, the Ruseckas model achieved a top-1 accuracy of 54.8%—the most challenging scenario—while the Batool model reached 46.2%. In contrast, the WiSARD model attained 58.2% top-1 accuracy, demonstrating its competitiveness among deep learning-based solutions.

Beyond accuracy, the training efficiency of the WiSARD network stands out. The WiSARD model required only 0.217 s for training, while the Ruseckas and Batool models required 2187.74 s and 1310.27 s, respectively. Another indicator of the complexity of a model is the number of parameters. While the Batool model requires 8.6 MB, the Ruseckas model requires 3.93 MB of parameters. On the other hand, for the LiDAR input, the WiSARD model only needs to define the threshold variance

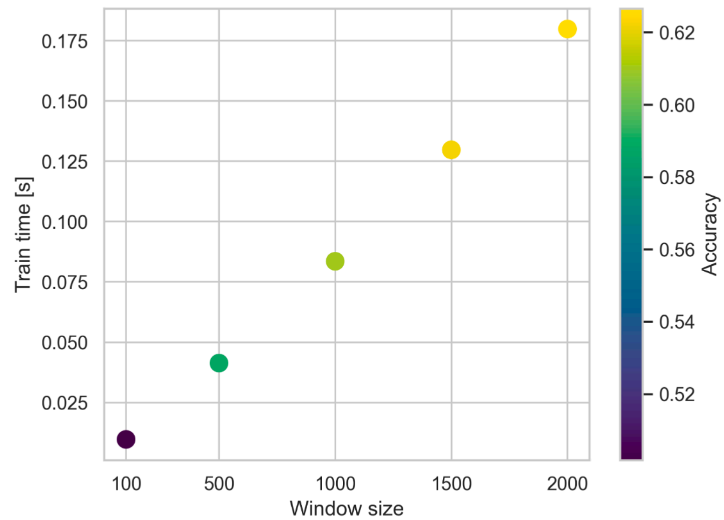

Fig. 8. Tradeoff metric: performance analyses of window sliding with 5 window size using LiDAR data.

and address size in memory. This result highlights WiSARD's ability to achieve strong performance with significantly lower computational cost.

Finally, it is important to emphasize that these results were obtained under batch learning conditions, where all training data are available simultaneously. This motivates the next step of our investigation: assessing how such models behave under a window-oriented (online) learning approach, which better reflects real-world communication dynamics and time-varying environments.

7.3. Performance WiSARD model

This subsection analyzes the WiSARD model's performance by first optimizing the window size for the sliding approach, and subsequently comparing the fixed, sliding, and incremental windowing strategies.

Having established the viability of the sliding window, we evaluated five different window sizes to determine the optimal configuration. The results, shown in Fig. 8, highlight the fundamental trade-off: larger window sizes improve accuracy by capturing more diverse environmental conditions but increase average training time. Conversely, smaller windows offer faster adaptation at the cost of accuracy. The training time distributions in Fig. 9 widen with larger windows; this is because they incorporate more data from the s008 dataset, which contains a higher average number of samples per episode (see Fig. 5).

These observations reinforce the importance of carefully balancing window size to optimize both accuracy and computational efficiency, depending on the real-time constraints of the application. To balance accuracy and computational efficiency, we applied a penalization function: $accuracy \cdot \beta - timetrain \cdot \alpha$.

The weighting coefficients were set to $\beta = 0.6$ and $\alpha = 0.4$ to reflect the specific requirements of robust beam management. We assign a higher weight to accuracy (β) because beam misalignment typically results in connection failure (outage), whereas training time (α) represents a latency cost that reduces system efficiency but does not necessarily break the link if kept within the channel coherence time. Sensitivity analysis confirms that the selection of the 1000-episode window as the optimal configuration is robust to modest variations in these parameters. This window size consistently maximizes the score by achieving high accuracy without incurring the excessive computational penalty associated with larger accumulated datasets

Fig. 10 presents the accuracy comparison for the three window types. As expected, the incremental window achieved the highest accuracy, as the model learns from a complete history of the environment's dynamics. However, this approach is infeasible for practical applications due to

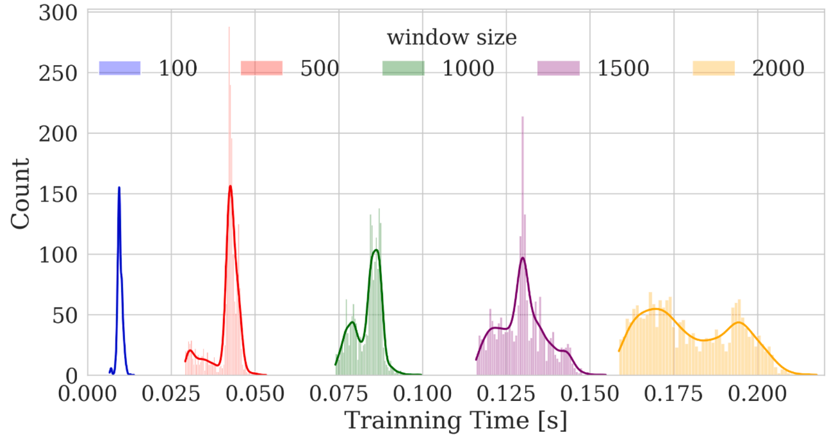

Fig. 9. Training time distributions of all sliding window sizes using LiDAR data.

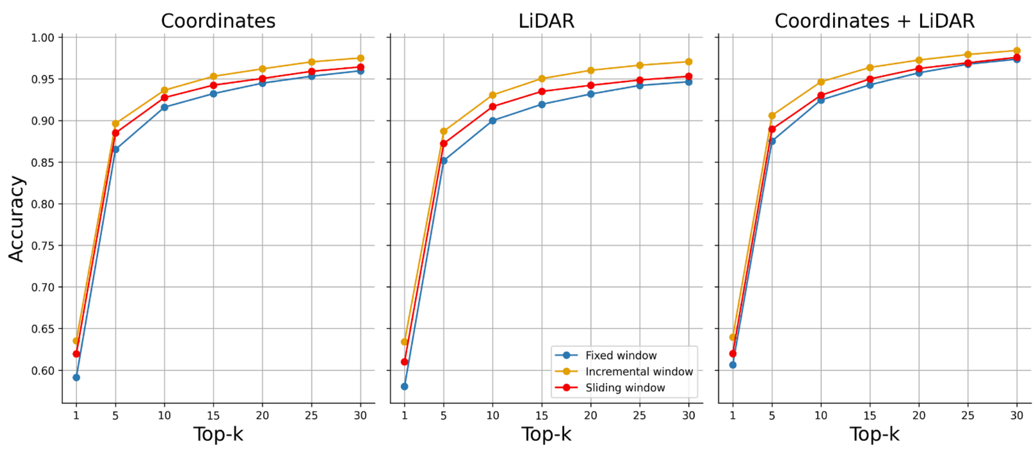

Fig. 10. performance comparison of WiSARD model between all types of windows.

unbounded storage and processing requirements. The fixed-window approach, which served as a baseline equivalent to batch training, demonstrated the lowest performance as the model was never updated. The sliding-window strategy yielded an intermediate accuracy, providing a balance between adaptation and historical data volume.

A notable gap was observed between Top-1 and Top-5 performance across all configurations (e.g., $\approx 61\%$ Top-1 vs. $> 90\%$ Top-5). This discrepancy offers two critical insights into the behavior of the model in dynamic mmWave environments:

1. *Angular Misalignment Sensitivity:* Due to the narrow beamwidth associated with the 32-element antenna array, small spatial drifts or GPS positioning errors can cause the optimal signal to shift to an immediate neighbor beam. While the Top-1 metric penalizes this as a complete error, the Top-5 metric successfully captures these slightly misaligned but viable links.
2. *Multipath Ambiguity:* The propagation channel frequently exhibits multiple strong clusters. In scenarios where a secondary path (e.g., a strong NLoS reflection) has a signal strength comparable to the primary path, the classifier may prioritize the secondary beam. Although technically suboptimal, these beams often provide sufficient connectivity, which is reflected in the high Top-5 accuracy.

Consequently, the high Top-5 scores confirm that the WiSARD model consistently identifies the correct spatial sector, even when the exact peak alignment is impacted by high-mobility dynamics.

The computational cost of each strategy is illustrated in Fig. 11. The incremental window's training time increases linearly as the dataset grows, while the fixed window's processing time remains constant. The sliding window (set at 1000 episodes) maintains a consistent average training time. The observed variance in its time distribution is a direct reflection of the variable number of samples contained within each episode, as shown in Fig. 5. These results confirm that the sliding window is a promising strategy, balancing high accuracy with efficient training.

7.4. Performance comparison between WiSARD and baseline models

The performance of the WiSARD model was benchmarked against two CNN-based baseline models from [8] and [22]. Fig. 12 presents the comparative results for the fixed, incremental, and sliding window adaptation strategies, evaluated using GPS, LiDAR, and combined LiDAR + GPS input modalities.

The results align with theoretical expectations for the different windowing strategies. The incremental window, which utilizes all historical data, consistently yielded the highest accuracy across all models and input types. Conversely, the fixed window, representing a static and limited knowledge of the environment, produced the lowest accuracy. The sliding window approach demonstrated an effective balance, allowing the models to adapt to environmental changes. A sliding window of 1000 episodes was found to provide an optimal trade-off, achieving high accuracy while using 47.96% less training data than the incremental approach.

Focusing on this optimal sliding window configuration, the simulation results show that the WiSARD model achieved the highest top-1 accuracy. It outperformed the best-performing baseline by 1.9% with

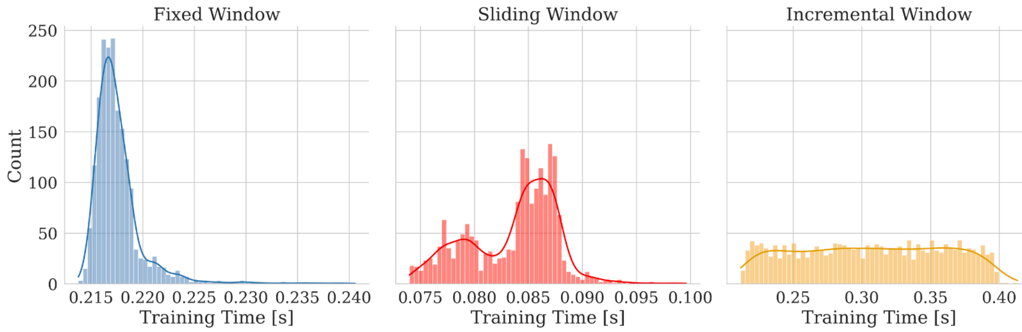

Fig. 11. Training time histogram of windows type.

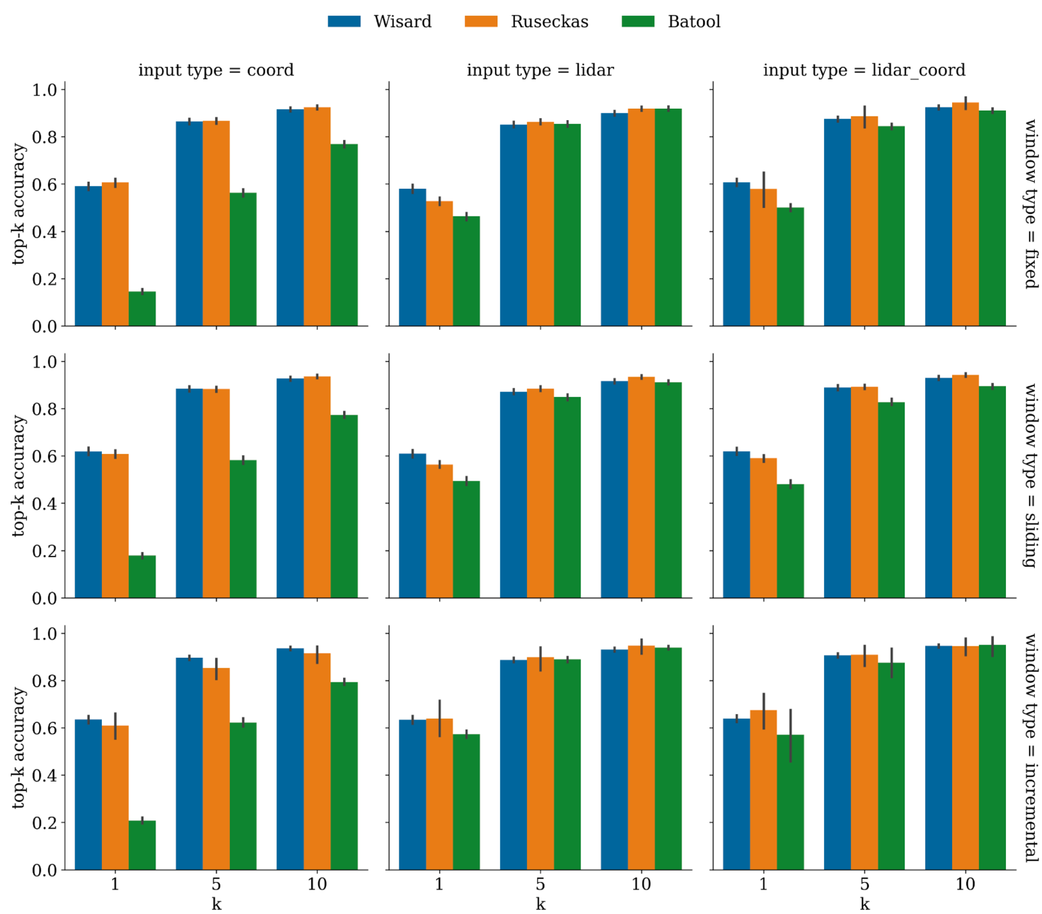

Fig. 12. Performance comparison of WiSARD and baseline models.

coordinates data, 4.6% with LiDAR data, and 3.0% with combined LiDAR + coordinates data.

Furthermore, Fig. 13 illustrates the comparative training time, in seconds, for the WiSARD against the baseline methods for the sliding window approach.

The results demonstrate a reduction in computational overhead of approximately three orders of magnitude. While the deep learning baselines require iterative backpropagation and floating-point matrix multiplications, the WiSARD framework relies exclusively on integer-based RAM addressing and binary write operations.

A logarithmic scale is employed for the Y-axis to properly accommodate the wide dynamic range of the observed results. Both baseline methods, [8] and [22], demonstrate substantial computational overhead, with training times registering in excess of 10^2 s with LiDAR data. In stark contrast, the proposed method achieves a significantly reduced training time, at less than 10^{-1} s.

This dramatic reduction in complexity is consistent with the efficiency goals highlighted in recent mmWave literature. As quantified by López-Benítez and Alhulayil [37] in the context of fading models, high computational burdens in channel estimation can severely bottleneck real-time system performance. We observe a parallel constraint in beam selection: complex neural architectures, despite their accuracy, introduce latency penalties that are incompatible with the rapid coherence times of 6G channels. By shifting the computational paradigm

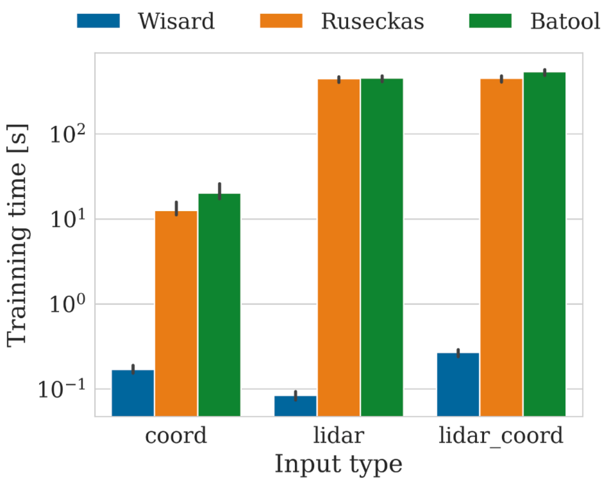

Fig. 13. Training time comparison of WiSARD and baseline model in sliding window.

from heavy matrix operations to lightweight memory lookups, the proposed framework satisfies these strict efficiency requirements, making it uniquely suitable for deployment on resource-constrained edge devices.

All experiments were executed on a virtual machine running Ubuntu 22.04.3 LTS (kernel version 5.15.0-124-generic), equipped with 8 CPU cores and 64 GB of RAM, ensuring consistency and reproducibility.

8. Conclusions and future works

In this paper, we proposed a window-based learning approach for mmWave beam selection utilizing a Weightless Neural Network (WNN), specifically the WiSARD model, informed by unimodal (GPS or LiDAR) and multimodal (GPS + LiDAR) contextual data. The results demonstrated that preprocessing techniques, namely thermometer encoding and variance reduction, were effective for applying WiSARD to the beam selection problem.

The WiSARD model achieved competitive performance compared to DNN-based techniques; however, its inherently low computational complexity makes it a more appropriate solution for hardware-restricted environments such as edge devices. Furthermore, the window-based learning mechanism joint to WNN demonstrated a high capacity for adaptation to scenario variations, yielding in the best case a 4.6% increase in top-1 accuracy while requiring 47% less training data than a conventional batch-based approach. This work provides an effective intermediate solution between static offline learning and fully online adaptation. As future work, we suggest the evaluation of efficient, fully online WNN algorithms designed to maintain real-time responsiveness, as the integration of such algorithms with contextual data presents a pivotal technique for achieving robust, high-throughput communication in next-generation wireless networks.

CRediT authorship contribution statement

Joanna C. Manjarres: Writing– review & editing, Writing – original draft, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization; **Douglas O. Cardoso:** Writing – review & editing, Validation, Supervision, Methodology, Conceptualization; **José Ferreira De Rezende:** Writing – review & editing, Validation, Resources, Project administration, Methodology, Conceptualization.

Data availability

https://www.lasse.ufpa.br/raymobtime

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Algorithms

Input: Batch of coordinates $\{(x_i, y_i)\}_{i=1}^N$, Scaling factor α, **Global Bounds** $X_{\min}, X_{\max}, Y_{\min}, Y_{\max}$
Output: Binary encoded vectors $\{\mathbf{e}_i\}_{i=1}^N$

1. **Initialize thermometer sizes (Fixed):**
$L_x \leftarrow (X_{\max} - X_{\min}) \times \alpha$; $\quad L_y \leftarrow (Y_{\max} - Y_{\min}) \times \alpha$.
2. **Encode each coordinate pair:**
for $i \leftarrow 1$ ***to*** N **do**
    `// Normalize using Global Bounds`
    $\Delta x_i \leftarrow x_i - X_{\min}$; $\quad \Delta y_i \leftarrow y_i - Y_{\min}$;
    Initialize $\mathbf{e}_{\mathbf{x_i}} \in \{0,1\}^{L_x}$, $\mathbf{e}_{\mathbf{y_i}} \in \{0,1\}^{L_y}$ (zeros);
    Set first $(\Delta x_i \times \alpha)$ bits of $\mathbf{e}_{\mathbf{x_i}}$ to 1;
    Set first $(\Delta y_i \times \alpha)$ bits of $\mathbf{e}_{\mathbf{y_i}}$ to 1;
    Concatenate: $\mathbf{e_i} \leftarrow [\mathbf{e}_{\mathbf{x_i}} \,||\, \mathbf{e}_{\mathbf{y_i}}]$.
end
return $\{\mathbf{e_i}\}_{i=1}^N$

Algorithm 1: Thermometer encoding for coordinate data.

Input: Data Stream $\mathcal{D}$ (Concatenation of s008 and s009), Transition Point T_{split}, Window Size W_{size}
Output: Accuracy metrics over time

```
Initialize Model M (Empty);
Initialize Window Buffer B ← ∅;
for each batch/episode E_t in D do
    // 1. Update Window and Retrain
    if Strategy == Fixed then
        if B == ∅ then
            B ← Load Static Set (s008);
        end
    end
    else if Strategy == Sliding then
        Add E_t to Buffer B;
        if |B| > W_size then
            Remove oldest batch from B;
        end
    end
    else if Strategy == Incremental then
        Add E_t to Buffer B;
    end
    M.train(B);
    // 2. Prediction Phase (Only on Test Data - s009)
    if t > T_split then
        Y_pred ← M.predict(E_t.features);
        Record Accuracy (Y_pred, E_t.labels);
    end
end
```

Algorithm 2: Online window-based evaluation protocol.

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